



Stock Price Prediction

Predict future stock prices using a simple Linear Regression model.

This prediction is based on historical closing prices and is intended for educational and portfolio purposes only.

Enter Stock Symbol

Training Start Date

Days to Predict

AAPL

2021/06/27

31

Historical stock data loaded successfully.

Historical Closing Prices

	Date	Close
1245	2026-06-12 00:00:00-04:00	291.13
1246	2026-06-15 00:00:00-04:00	296.42
1247	2026-06-16 00:00:00-04:00	299.24
1248	2026-06-17 00:00:00-04:00	295.95
1249	2026-06-18 00:00:00-04:00	298.01
1250	2026-06-22 00:00:00-04:00	297.01
1251	2026-06-23 00:00:00-04:00	294.3
1252	2026-06-24 00:00:00-04:00	293.08
1253	2026-06-25 00:00:00-04:00	275.15
1254	2026-06-26 00:00:00-04:00	283.78

⚙️ Feature Engineering

The model uses the day number as the input feature and Closing Price as the target variable.

	Date	Close	Day
1250	2026-06-22 00:00:00-04:00		297.011250
1251	2026-06-23 00:00:00-04:00		294.31251
1252	2026-06-24 00:00:00-04:00		293.081252
1253	2026-06-25 00:00:00-04:00		275.151253
1254	2026-06-26 00:00:00-04:00		283.781254

✅ Linear Regression Model Trained Successfully

📊 Model Performance

R² Score

0.8074

Training Samples

1255

R² Score measures how well the Linear Regression model fits the historical data.

- Closer to 1.0 → Better Fit
- Closer to 0.0 → Poor Fit

Since this project uses a simple Linear Regression model, the score is intended only for educational demonstration.

🔮 Future Stock Price Prediction

Future stock prices predicted successfully.

	Date	Predicted Closing Price
0	2026-06-29 00:00:00-04:00	263.0165
1	2026-06-30 00:00:00-04:00	263.1273
2	2026-07-01 00:00:00-04:00	263.2381
3	2026-07-02 00:00:00-04:00	263.3489
4	2026-07-03 00:00:00-04:00	263.4597
5	2026-07-06 00:00:00-04:00	263.5705
6	2026-07-07 00:00:00-04:00	263.6813
7	2026-07-08 00:00:00-04:00	263.7921
8	2026-07-09 00:00:00-04:00	263.903
9	2026-07-10 00:00:00-04:00	264.0138



Historical vs Predicted Price



Prediction Insights

Current Price

\$283.78

Predicted Price (31 Days)

\$266.34

Expected Return

-6.15%

🧠 Model Interpretation



Moderate Bearish Trend

The model predicts a moderate downward trend.

Investors should monitor price action before taking decisions.



Investment Summary

Metric	Value
Current Price	\$283.78
Predicted Price	\$266.23
Expected Return	-6.18%
Model Accuracy (R^2)	0.8074
Recommendation	🔴 REDUCE



Quick Analysis

Current Price

\$283.78

Expected Return

-6.18%

Predicted Price

\$266.23

Recommendation

🔴 REDUCE

About This Prediction Model

Machine Learning Model Used

This project uses a **Linear Regression** model from the Scikit-Learn library.

The model learns the historical closing price trend and estimates future prices using a simple linear relationship between time and stock price.

This model is intentionally simple because the goal of this project is to demonstrate:

- Data Collection using Yahoo Finance
- Data Cleaning using Pandas
- Feature Engineering
- Machine Learning with Scikit-Learn
- Data Visualization using Plotly
- Interactive Dashboard using Streamlit

Model Limitations

The prediction generated by this dashboard is based only on historical closing prices.

The model DOES NOT consider:

- Company Financial Results
- Quarterly Earnings
- Market News
- Global Events
- Interest Rates
- Investor Sentiment
- Government Policies

Therefore, actual market prices may differ significantly from the predicted values.